

Ish Thumber

ishthumber343@gmail.com | +91 9909919803 | ishthumber.vercel.app | linkedin.com/in/ishthumber | github.com/IshThumber
medium.com/@ishthumber

SUMMARY

Cloud Engineer (Python, FastAPI, React) specialising in AI-driven systems. Delivers significant efficiency gains by accelerating development cycles and optimising cloud environment setup.

PROFESSIONAL EXPERIENCE

Cloud Engineer

March 2024 – Present | Hyderabad, India

Searce Inc.

- Designed reusable, modular Terraform IaC components that standardised environment provisioning across multiple client projects, reducing infrastructure duplication by 30% and accelerating team onboarding efficiency.
- Migrated large-scale GCP Apigee infrastructure using Terraform, implementing Private Service Connect (PSC) architecture to replace legacy systems and enable seamless multi-environment scalability.
- Engineered a cross-region disaster recovery strategy (Mumbai to Hyderabad) using AWS Lambda and EC2 Data Lifecycle Manager with RDS Point-In-Time Recovery, achieving a 15-minute RTO for critical workloads.
- Implemented an observability stack using Prometheus, Grafana, Loki, and Alertmanager to monitor 150+ EC2 instances with custom alert rules, reducing MTTR by 40%.
- Designed high availability Fortinet firewall architectures on AWS with automated failover mechanisms, achieving 99.99% uptime for enterprise security workloads.
- Deployed production-grade Kubernetes monitoring on Amazon EKS with Prometheus, Grafana, and Alertmanager for automated incident notifications while implementing RBAC policies to strengthen cluster security.

SKILLS

- **Programming Languages:** Python, JavaScript, C++
- **Cloud Platforms:** Amazon Web Services (EC2, S3, IAM, VPC), Microsoft Azure (Azure Blob Storage, Azure Virtual Machines), Google Cloud Platform (Google Cloud Storage, Compute Engine)
- **IaC & Automation:** Terraform, Helm, Jenkins, GitHub Actions, AWS CodePipeline, AWS CodeCommit, AWS CodeBuild
- **Monitoring & Logging:** Prometheus, Grafana, Loki, AlertManager, CloudWatch
- **Containerisation & Orchestration:** Docker, Kubernetes, Amazon Elastic Kubernetes Service
- **Web Development:** React.js, Node.js, Express.js, Tailwind CSS, HTML, CSS
- **Core Subjects:** Data Structures and Algorithms, Cloud Computing, Operating Systems, Computer Networks, Database Management Systems, Software Engineering, Object-Oriented Programming

PROJECTS

Observability Accelerator

September 2025 – Present

- Reduced monitoring setup time from hours to under 10 minutes by architecting a React 19 guided deployment wizard with Server Actions and use hooks, eliminating manual configuration errors across 20+ microservices and improving developer productivity organisation-wide.
- Implemented real-time deployment visibility via Server-Sent Events (SSE), streaming 20+ pipeline stage updates to a terminal-style UI with execution timing, stage-level status, and actionable failure diagnostics for complex async workflows.
- Architected an AWS metrics integration layer featuring CloudWatch metrics selection with tag-based filtering and automatic service discovery (EC2, RDS), enabling seamless ingestion of AWS-native telemetry into Prometheus without manual exporter configuration.
- Delivered unified observability dashboards combining metrics (Prometheus), logs (Loki), traces (Tempo), and CloudWatch with preconfigured alerts and multi-channel notifications to accelerate incident response and reduce MTTR.
- Orchestrated end-to-end full-stack workflows handling async state, retries, and partial failure recovery across backend services, while implementing optimistic UI patterns to maintain 100% state accuracy during long-running infra provisioning.
- Established CI/CD pipelines and end-to-end testing frameworks to automate deployment validation, ensuring reliable delivery of the complete observability stack from development to production.

RAG Flashcards Generator

August 2025 – September 2025

- Developed a scalable, serverless backend using FastAPI with asynchronous I/O and modular services, deployed on AWS Lambda and API Gateway.
- Engineered a complete Retrieval-Augmented Generation (RAG) pipeline, achieving sub-second semantic search by integrating AWS Textract, Amazon Titan embeddings (1024-dim), and OpenSearch kNN.
- Incorporated Claude 3.5 Sonnet for generating context-aware flashcards, ensuring high factual accuracy through citation tracking and retrieval grounding.

CERTIFICATIONS

AWS Certified Solutions Architect – Professional (SAP-C02), AWS Certified Solutions Architect – Associate (SAA-C03), Terraform Certified Associate (HCTAO-003), AWS Certified Cloud Practitioner (CLF-02)

RECOGNITION

AWS Community Builder – DevTools

AWS

Recognized by AWS for contributions to cloud automation, DevOps, and infrastructure development using tools like CodePipeline, Terraform, and Kubernetes.

EDUCATION

Bachelors in Technology, Computer Engineering

July 2020 – May 2024

Charotar University of Science and Technology

CGPA: 8.5